

Credit Card Spending Behavior Analysis

Business Intelligence Report

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Role: Data Analyst

Tools: Excel | Python | MySQL | SQL | Tableau

Domain: Banking & Financial Services

Business Background

Credit cards have become one of the primary payment methods in modern banking. Every transaction generates valuable information that helps banks understand customer spending habits, identify profitable segments, improve retention, optimize reward programs, detect fraud, and maximize revenue. This project transforms raw transaction data into actionable business intelligence using Excel, Python, MySQL, SQL, and Tableau.

Business Problem

The bank wants to identify its highest-value customers, understand which card types and merchant categories generate the most revenue, evaluate payment preferences, analyze income and age group behavior, monitor fraud, and make informed strategic decisions.

Project Objective

Analyze customer transaction data and convert it into actionable insights that support better business decisions and improve profitability.

Dataset Overview

- Customers: 3,001
- Transactions: 5,000
- Total Spending: █47.59 Million
- Average Transaction: █9,518
- Fraud Rate: 4.86%

Tools Used

Excel – Data Cleaning

Python – Data Cleaning & EDA

MySQL – Database Management

SQL – Business Analysis

Tableau – Interactive Dashboard

Dashboard Analysis

The executive dashboard provides a complete view of customer spending behavior. Total spending of █47.59M indicates healthy card utilization. Monthly spending remains stable between █3.5M and █4.3M, suggesting predictable revenue. The Top 10 Customers contribute a significant share of revenue and should be retained through loyalty programs. Electronics (█14.98M) and Travel (█10.76M) are the highest spending categories, creating opportunities for merchant partnerships. Silver cards generate the highest spending because they represent the largest customer base. High-income customers contribute █26.07M, making them the most valuable customer segment. Payment methods are evenly distributed across Chip, Tap, Swipe, and Online, showing balanced customer preferences. Fraud transactions remain low but require continuous monitoring.

Key Business Insights

1. Stable monthly spending indicates consistent customer engagement.
2. Electronics and Travel dominate transaction value.
3. High-income customers generate the largest share of revenue.
4. Silver cardholders offer strong upgrade potential to Gold and Platinum.
5. A small group of customers contributes disproportionately to total revenue.

6. Digital payment adoption is healthy.
7. Fraud remains manageable but must be continuously monitored.

Strategic Recommendations

- Launch premium card upgrade campaigns.
- Build partnerships with electronics retailers, airlines, and hotels.
- Introduce VIP loyalty programs for high-value customers.
- Promote digital payments with cashback and rewards.
- Deploy AI-based fraud detection and instant customer alerts.
- Personalize offers based on customer age, income, and spending behavior.

Conclusion

This project demonstrates end-to-end analytics using Excel, Python, MySQL, SQL, and Tableau. The analysis converts raw transaction data into business intelligence that supports strategic decision-making, improves customer experience, strengthens fraud prevention, and identifies opportunities for sustainable revenue growth.

Skills Demonstrated

Data Cleaning • Exploratory Data Analysis • SQL Queries • Database Design • Tableau Dashboard Development • KPI Reporting • Business Storytelling • Banking Domain Analytics • Strategic Decision Support